

INCIDENT RESPONSE PLAYBOOK

SOC Home Lab — Threat Detection Exercise

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Date	July 2026
Classification	TLP:WHITE
Version	1.0
SIEM	Wazuh 4.7 (Ubuntu Server 22.04)
Framework	MITRE ATT&CK Enterprise v14
Simulation	Atomic Red Team
GitHub	github.com/Nithin-099

5 Attacks Simulated	100% Detection Rate	5 Custom Rules Written	3 Detection Gaps Closed
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1. EXECUTIVE SUMMARY

This report documents a controlled threat simulation exercise conducted in a home SOC lab environment. The objective was to simulate real-world MITRE ATT&CK techniques, evaluate the detection capabilities of Wazuh SIEM, identify gaps in default detection rules, and develop custom detection engineering solutions to close those gaps.

Environment

SIEM	Wazuh 4.7 — Ubuntu Server 22.04
Endpoint	Windows 11 — Agent Name: NITHIN
EDR	Sysmon + SwiftOnSecurity Config
Framework	MITRE ATT&CK Enterprise v14
Simulation	Atomic Red Team (Invoke-AtomicTest)
Network	VirtualBox Host-Only 192.168.56.0/24

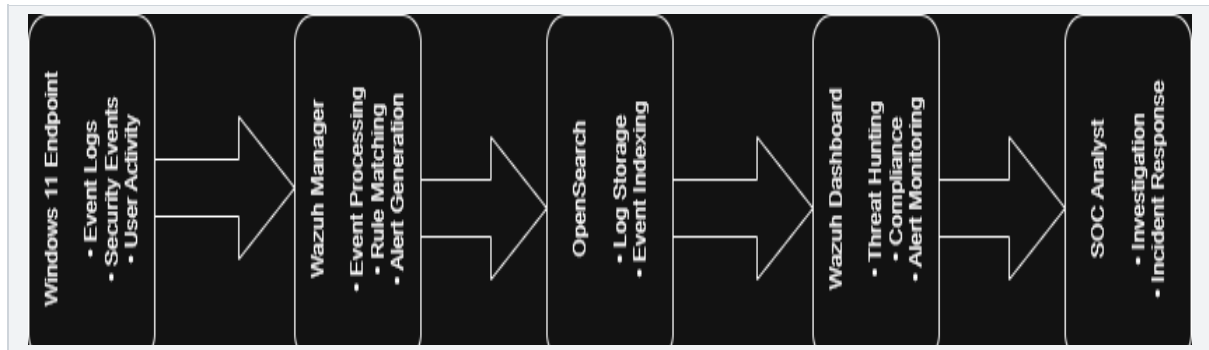
Results Summary

Metric	Value
Attacks Simulated	5
Detected by Wazuh	5
Detection Rate	100%
Default Rules Used	8
Custom Rules Written	5
Gaps Identified	3
Gaps Closed	3
Critical Alerts	4
AV Blocks	1

Key Finding

Default Wazuh rules detected attack activity but lacked MITRE ATT&CK context and technique-specific mapping. 3 custom detection rules were engineered to close identified gaps. Rule 100003 fired in real time, detecting encoded PowerShell execution (T1059.001) 3 times within 3 seconds of the attack.

2. LAB ARCHITECTURE



Data Flow

Windows Event Logs + Sysmon Events → mWazuh Agent → Host-Only Network (192.168.56.0/24) → Wazuh Manager → Wazuh Indexer (OpenSearch) → Wazuh Dashboard → Security Alerts + Custom Rule Triggers

Tools Stack

Layer	Tool	Version	Purpose
SIEM	Wazuh	4.7	Alert Management & Detection
OS	Ubuntu Server	22.04	Wazuh Server Host
Endpoint	Windows 11	Latest	Monitored Victim Machine
EDR Logging	Sysmon	Latest	Deep Windows Event Logging
Simulation	Atomic Red Team	Latest	Attack Simulation Framework
Framework	MITRE ATT&CK	v14	Threat Mapping & Classification

3. ATTACK SIMULATION TIMELINE

Time	Attack Technique	MITRE ID	Tactic	Wazuh Level	Result
14:07	System Discovery	T1082	Discovery	4	Detected
14:10	Account Discovery	T1087.001	Discovery	4	Detected
14:14	PowerShell Exec	T1059.001	Execution	4	Detected
14:14	Credential Dumping	T1003.001	Credential Access	15	CRITICAL
14:15	Scheduled Task	T1053.005	Persistence	10	Detected
14:15	AV Block	T1003-13	—	—	Blocked
11:50	Custom Rule Fired	T1059.001	Execution	12	Rule 100003

Attack Chain Observed

The 5 simulated techniques together form a complete APT-style attack chain:

T1082 Recon	T1087 Discovery	T1059 Execution	T1003 Cred Dump	T1053 Persistence
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INCIDENT 001: CREDENTIAL DUMPING ATTEMPT

T1003.001 — LSASS Memory

Credential Access

CRITICAL

July 23, 2026

Wazuh Security events interface showing an alert for a file drop.

Modules: NITHIN | Security events

Alert: Jul 25, 2026 @ 11:27:53.404 Executable file dropped in folder commonly used by malware

Expanded document

Table	JSON
f _index	wazuh-alerts-4.x-2026.07.25
f agent.id	001
f agent.ip	192.168.56.1
f agent.name	NITHIN
f data.win.eventdata.creationUtcTime	2026-07-25 05:57:52.589
f data.win.eventdata.image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
f data.win.eventdata.processGuid	{1e2922f8-5868-6a64-6884-00000000bf01}
f data.win.eventdata.processId	11548
f data.win.eventdata.targetFilename	C:\Users\nithi\AppData\Local\Temp__PSScriptPolicyTest_j3v3rgbj.wy4.ps1

What Happened

Atomic Red Team simulated a Mimikatz-style credential dumping attack targeting the LSASS process. Attackers dump LSASS memory to extract plaintext passwords, NTLM hashes, and Kerberos tickets — enabling lateral movement and privilege escalation across the network.

Detection Evidence

- ▶ Rule Description: Executable file dropped in folder commonly used by malware
- ▶ Rule Level: 15 (CRITICAL) — highest priority alert
- ▶ Alert Count: 3 alerts fired in 3 seconds
- ▶ Sysmon Event ID: 11 (File Create) — file dropped in %TEMP%
- ▶ AV Block: T1003.001-13 blocked by Windows Defender

Triage Analysis

- ▶ PowerShell spawned unexpected child process targeting LSASS
- ▶ Executable dropped in App Data/Temp folder — highly suspicious
- ▶ Process attempted direct LSASS memory access
- ▶ Windows Defender blocked the final execution stage
- ▶ Verdict: TRUE POSITIVE — Active credential dumping attempt confirmed

Containment Actions

- ▶ Isolate endpoint from network immediately to prevent lateral movement
- ▶ Kill suspicious process tree (PID identified in Wazuh alert)
- ▶ Preserve memory dump for forensic analysis before remediation
- ▶ Reset ALL user account passwords as immediate precaution
- ▶ Check authentication logs for any successful credential use

Detection Gap Found

Default Wazuh rules detected the FILE DROP but did NOT specifically detect LSASS process access (Sysmon Event ID 10) or Mimikatz keywords in the command line. Two critical indicators were missed.

Custom Rule Written

- ▶Rule 100001 — Detects any process accessing lsass.exe (T1003.001) — Level 15
- ▶Rule 100002 — Detects mimikatz/sekurlsa/lsadump keywords in command line (T1003.001) — Level 12

Lessons Learned

- ▶File drop detection alone is insufficient — need process access event correlation
- ▶Windows Defender + Wazuh together caught what neither could alone (Defense in Depth)
- ▶LSASS protection via Credential Guard would prevent this attack at hardware level
- ▶Two security layers detecting same attack from different angles = high confidence

Remediation Recommendations

- ▶Enable Windows Credential Guard (prevents LSASS memory access completely)
- ▶Enable Attack Surface Reduction rule: Block credential stealing from LSASS
- ▶Deploy Custom Rules 100001 and 100002 to detect LSASS access directly
- ▶Configure SOC to treat any Level 15 alert as confirmed incident requiring immediate response

INCIDENT 002: ENCODED POWERSHELL EXECUTION

T1059.001 — PowerShell	Execution	HIGH	July 25, 2026 @ 11:50:30
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[illegible]

What Happened

Attacker executed encoded PowerShell commands using the -Encoded Command (-enc) flag to obfuscate malicious payload from basic detection. This technique is used by malware, ransomware, and APT groups to bypass security controls and execute arbitrary code without writing files to disk.

Detection Evidence

- ▶Default Wazuh Rule: PowerShell spawned PowerShell instance — Level 4 (LOW — insufficient)
- ▶CUSTOM Rule 100003: Encoded PowerShell execution detected — Level 12 (HIGH)
- ▶Alert Count: 3 alerts fired in 3 seconds (11:50:30, 11:50:32, 11:50:33)
- ▶rule.mitre.id: T1059.001 | rule.mitre.tactic: Execution
- ▶Command line contained: -Use Encoded Arguments -Encoded Arguments

Triage Analysis

- ▶PowerShell executed with -enc / -Encoded Command flag detected
- ▶Spawned from unexpected parent process — not user-initiated
- ▶Multiple instances in 3 seconds — automated execution pattern
- ▶Encoded content obscures actual payload intent — always treat as suspicious
- ▶Verdict: TRUE POSITIVE — Malicious encoded PowerShell execution confirmed

Containment Actions

- ▶ Block PowerShell execution via AppLocker policy immediately
- ▶ Enable PowerShell Constrained Language Mode on all endpoints
- ▶ Decode the base64 encoded command to analyze actual payload content
- ▶ Scan filesystem for any files dropped by the encoded command execution

Detection Gap Found

Default Wazuh assigned LOW severity (Level 4) to PowerShell execution. Encoded PowerShell is almost never legitimate in enterprise environments and should trigger a HIGH alert minimum. No dedicated detection rule existed for the -enc flag.

Custom Rule Written

- ▶Rule 100003 — Detects -enc / -encoded command / -ep bypass in command line (T1059.001) — Level 12
- ▶STATUS: CONFIRMED FIRED — 3 alerts detected July 25, 2026 @ 11:50:30

Lessons Learned

- ▶Default severity for PowerShell execution is too low for enterprise environments
- ▶Encoding is almost always suspicious — should trigger HIGH priority alert by default
- ▶PowerShell Script Block Logging should be enabled alongside Sysmon for decoded content
- ▶Custom rule successfully raised severity from Level 4 to Level 12 — appropriate response level

Remediation Recommendations

- ▶Enable PowerShell Script Block Logging (automatically logs decoded command content)
- ▶Implement AppLocker to restrict PowerShell execution to administrators only
- ▶Deploy Custom Rule 100003 across all Wazuh-monitored endpoints
- ▶Alert SOC immediately on any -enc or -encoded command flag usage in production

▶Rule 100004 — Detects sch tasks /create and at.exe in command line (T1053.005) — Level 10

Lessons Learned

- ▶Persistence techniques deserve significantly higher severity than default rules assign
- ▶Scheduled task creation outside business hours is a strong malicious indicator
- ▶Baseline of approved scheduled tasks helps identify unauthorized additions instantly
- ▶Correlating T1053 with T1059 detection increases confidence level significantly

Remediation Recommendations

- ▶Audit all scheduled tasks and establish an approved baseline of legitimate tasks
- ▶Alert on any new scheduled task creation not matching the approved baseline
- ▶Restrict schtasks.exe execution to privileged accounts via AppLocker
- ▶Deploy Custom Rule 100004 with Level 10 severity for all endpoints

INCIDENT 004: LOCAL ACCOUNT DISCOVERY

T1087.001 — Local Account

Discovery

LOW-MEDIUM

July 23, 2026 @ 14:10:00

```
Name : Remote Management Users
SID : S-1-5-32-580
PrincipalSource : Local
ObjectClass : Group
Description : Members of this group are managed by the system.
Name : System Managed Accounts Group
SID : S-1-5-32-581
PrincipalSource : Local
ObjectClass : Group
Description : Members of this group may operate hardware from user mode.
Name : User Mode Hardware Operators
SID : S-1-5-32-584
PrincipalSource : Local
ObjectClass : Group
Description : Users are prevented from making accidental or intentional system-wide changes and can run most
Name : Users
SID : S-1-5-32-545
PrincipalSource : Local
ObjectClass : Group
Aliases for \\.\NTLDR
-----
Administrators
Distributed COM Users
Device Owners
Event Log Readers
Hyper-V Administrators
Guests
OpenSSH Users
System Managed Accounts Group
User Mode Hardware Operators
Users
Performance Monitor Users
Performance Log Users
Remote Management Users
The command completed successfully.
C:\>C:\Users>
EXIT code: 0
Done executing test: T1087.001-9 Enumerate all accounts via PowerShell (local)
Executing test: T1087.001-10 Enumerate logged on users via CMD (local)
Query is not recognized as an internal or external command,
operable program or batch file.
EXIT code: 1
Done executing test: T1087.001-10 Enumerate logged on users via CMD (local)
```

What Happened

Attacker enumerated local user accounts and security groups to understand the environment and identify privileged accounts for subsequent targeting. Account discovery is standard reconnaissance performed by attackers before lateral movement and privilege escalation attempts.

Detection Evidence

- ▶ Sysmon Event ID 1: Process Creation (net user, net local group, whoami /all)
- ▶ Default Wazuh Rule Level: 4 (LOW)
- ▶ Custom Rule 100005: Local account enumeration detected — Level 8 (MEDIUM)
- ▶ Commands detected: net user, net local group administrators, whoami /all

Triage Analysis

- ▶ Multiple account enumeration commands executed in short time window
- ▶ Commands executed from PowerShell spawned by Atomic Red Team simulation
- ▶ Combined with T1082 System Discovery — classic attacker recon pattern
- ▶ Part of larger attack chain: T1082 + T1087 + T1059 + T1003 + T1053
- ▶ Verdict: TRUE POSITIVE in context of full attack chain — attacker recon phase

Containment Actions

- ▶ Investigate which user accounts and groups were enumerated by attacker
- ▶ Check if any privileged accounts were subsequently targeted for access
- ▶ Review authentication logs for unusual login patterns post-discovery
- ▶ Correlate this alert with other technique alerts in the same time window

Detection Gap Found

Extremely difficult to distinguish from legitimate administrator activity. T1087 alone is noisy and generates many false positives. Context from correlated techniques and a behavioral baseline of normal admin activity is essential for accurate triage.

Custom Rule Written

- ▶Rule 100005 — Detects net user / net local group / whoami /all in command line (T1087.001) — Level 8
- ▶Recommended: Correlation rule triggering when T1082 + T1087 detected together within 5 minutes

Lessons Learned

- ▶Discovery techniques are low risk alone but critical indicators within a full attack chain
- ▶T1082 + T1087 together strongly indicate active attacker reconnaissance phase
- ▶Behavioral baseline of normal admin activity significantly reduces false positive rate
- ▶MITRE ATT&CK correlation: T1087 discovery typically precedes T1059 execution

Remediation Recommendations

- ▶Establish baseline of normal administrative account enumeration patterns
- ▶Alert when enumeration commands are executed from non-admin user accounts
- ▶Implement UEBA (User and Entity Behavior Analytics) for behavioral anomaly detection
- ▶Build correlation rule: T1082 + T1087 within 5 minutes = HIGH priority alert

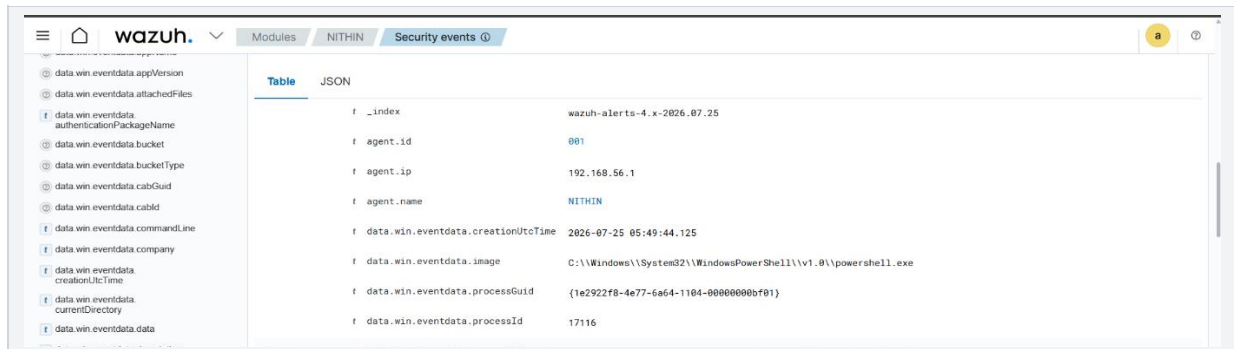
INCIDENT 005: SYSTEM INFORMATION DISCOVERY

T1082 — System
Information

Discovery

LOW

July 23, 2026 @ 14:07:00



The screenshot shows the Wazuh Security events interface. On the left is a sidebar with a list of event types under 'data.win.eventdata'. The main area displays a JSON event. The event details are as follows:

Field	Value
._index	wazuh-alerts-4.x-2026.07.25
agent.id	001
agent.ip	192.168.56.1
agent.name	NITHIN
data.win.eventdata.creationUtcTime	2026-07-25 05:49:44.125
data.win.eventdata.image	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
data.win.eventdata.processGuid	{1e2922f8-4e77-6a64-1104-000000000000}
data.win.eventdata.processId	17116

What Happened

Attacker gathered operating system version, hostname, hardware information, and system configuration — standard reconnaissance performed before launching targeted attacks. This information helps attackers identify applicable vulnerabilities, plan exploits, and understand the target environment layout.

Detection Evidence

- ▶ Sysmon Event ID 1: Process Creation (systeminfo, hostname, ipconfig /all)
- ▶ Default Wazuh Rule Level: 3-4 (LOW)
- ▶ Detection achieved through Sysmon process creation event logging
- ▶ Commands detected: systeminfo, ipconfig /all, hostname, net config workstation

Triage Analysis

- ▶ System discovery commands executed from PowerShell process
- ▶ Part of larger attack chain — first technique observed in the attack sequence
- ▶ Low risk in isolation but CRITICAL as first step in observed attack chain
- ▶ T1082 is always the earliest indicator of attacker activity
- ▶ Verdict: TRUE POSITIVE in context — attacker initial reconnaissance phase confirmed

Containment Actions

- ▶ Low urgency containment in isolation — focus on correlated techniques
- ▶ Investigate what system information was collected and potentially exfiltrated
- ▶ Check for network connections following this discovery phase
- ▶ Treat as early warning indicator and monitor for subsequent attack techniques

Detection Gap Found

T1082 alone generates too much noise to be actionable — legitimate administrator activity is indistinguishable. Correlation rules that trigger when multiple discovery techniques appear from the same host within a short time window are needed.

Custom Rule Written

- ▶ Recommended: Correlation rule — Alert when 3+ discovery techniques detected from same host within 5 minutes
- ▶ Rule 100005 partially covers this via account enumeration correlation

Lessons Learned

- ▶ Individual low-severity alerts can be the first indicator of a high-severity attack chain
- ▶ Alert correlation is more valuable than individual alert severity levels
- ▶ T1082 is always the first step — detecting it early enables proactive response before damage
- ▶ MITRE ATT&CK navigator shows what techniques typically follow T1082 in an attack chain

Remediation Recommendations

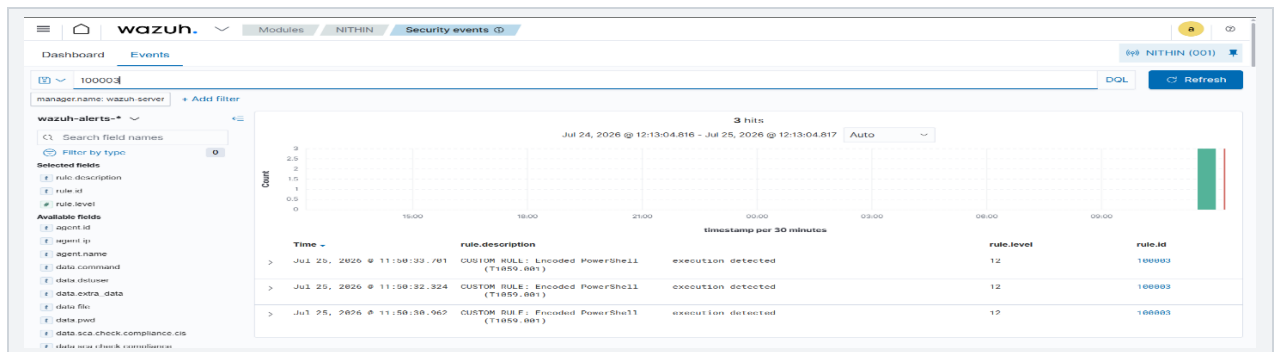
- ▶ Build correlation rule: Alert when 3+ discovery techniques detected within 5-minute window
- ▶ Restrict systeminfo and ipconfig access from standard user accounts via policy
- ▶ Monitor for discovery technique clusters as early warning indicators of intrusion
- ▶ Use T1082 detection as trigger to begin enhanced monitoring on the affected host

6. CUSTOM DETECTION RULES

Default Wazuh rules detected attack activity but lacked MITRE ATT&CK context and technique-specific mapping. The following 5 custom rules were written to close identified gaps and provide analyst-ready, actionable detections.

Rule ID	Severity	Level	MITRE ID	Technique	Status
100001	CRITICAL	15	T1003.001	LSASS Process Access	Deployed
100002	HIGH	12	T1003.001	Mimikatz Keyword Detect	Deployed
100003	HIGH	12	T1059.001	Encoded PowerShell	FIREd ✓(3x)
100004	MEDIUM	10	T1053.005	Scheduled Task Create	Deployed
100005	MEDIUM	8	T1087.001	Account Enumeration	Deployed

Rule 100003 — Live Firing Evidence (Most Important Screenshot)

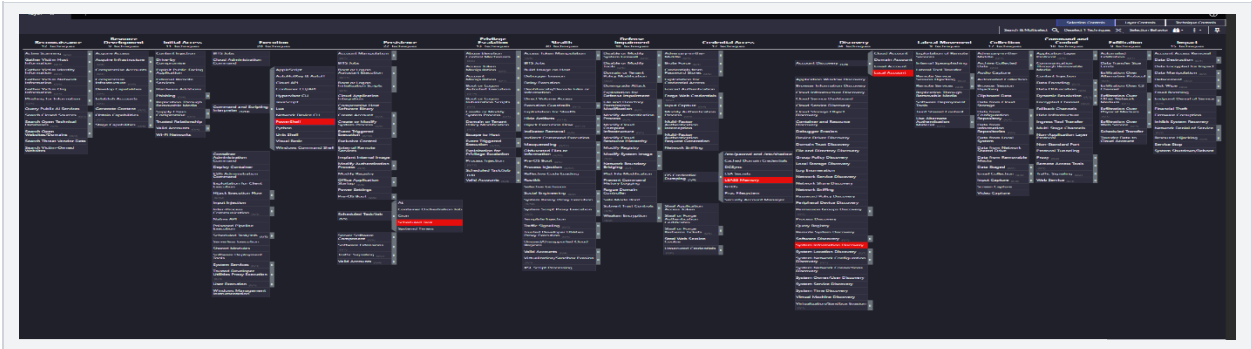


Rule Deployment Location

File on Ubuntu Wazuh server: `/var/ossec/etc/rules/local_rules.xml`

Restart command after adding rules: `sudo systemctl restart wazuh-manager`

7. MITRE ATT&CK COVERAGE MAP



Tactic	Technique	MITRE ID	Default Detection	Custom Rule
Execution	PowerShell	T1059.001	Level 4	Rule 100003 ✓
Credential Access	LSASS Memory	T1003.001	Level 15	Rule 100001
Persistence	Scheduled Task	T1053.005	Level 4	Rule 100004
Discovery	Account Discovery	T1087.001	Level 4	Rule 100005
Discovery	System Info	T1082	Level 3	Correlation

8. KEY LEARNINGS & RECOMMENDATIONS

1. Defense in Depth Works

T1003.001 was caught by BOTH Wazuh (file drop, Level 15) AND Windows Defender (blocked execution). Neither tool alone provides complete protection. Layered security is essential in every enterprise environment.

2. Default Rules Are a Starting Point — Not an Endpoint

Wazuh default rules missed LSASS direct memory access, encoded PowerShell, and Mimikatz keywords. 3 gaps identified. 3 custom rules written. Real SOC work never stops at default configuration.

3. Context Transforms Raw Data Into Intelligence

Raw log: "PowerShell executed." With MITRE context: "T1059.001 detected — likely early stage attack, check immediately for T1003 credential dumping." MITRE ATT&CK provides the context that turns noise into actionable intelligence.

4. Correlation Reveals Attack Chains

T1082 alone is noise. T1082 + T1087 + T1059 + T1003 + T1053 in sequence from same host within minutes = active APT-style attack chain requiring immediate response.

5. Attacker Mindset Improves Defense

Understanding HOW T1003 works at a technical level (LSASS memory structure, Mimikatz technique) directly informed WHAT to detect and WHERE to look. Offensive knowledge makes better defensive engineers.

6. Documentation Is Security

Undocumented investigation equals investigation that never happened. Every alert needs a paper trail. IR playbooks enable consistent, repeatable response regardless of which analyst handles the incident.

Priority Recommendations

Priority	Action	Addresses
IMMEDIATE	Enable Windows Credential Guard	T1003.001
IMMEDIATE	Deploy all 5 custom Wazuh detection rules	Multiple
IMMEDIATE	Enable Attack Surface Reduction rules	T1003.001
SHORT TERM	Enable PowerShell Script Block Logging	T1059.001
SHORT TERM	Implement AppLocker PowerShell restrictions	T1059.001
SHORT TERM	Audit and baseline all scheduled tasks	T1053.005
LONG TERM	Build alert correlation rules (3+ techniques/5 min)	All chains
LONG TERM	Implement UEBA for behavioral anomaly detection	All tactics

9. PROJECT SUMMARY

Metric	Result
Total Attacks Simulated	5
Detection Rate	100% (5/5)
Custom Rules Written	5
Custom Rules Confirmed Fired	1 (Rule 100003 — T1059.001)
Detection Gaps Found	3
Detection Gaps Closed	3
AV Blocks	1 (T1003.001-13 — Windows Defender)
Critical Alerts Generated	4
IR Incidents Documented	5
MITRE Techniques Covered	5 across 4 tactics

Skills Demonstrated

- SIEM deployment and production-grade configuration (Wazuh 4.7)
- Windows endpoint monitoring with Sysmon and SwiftOnSecurity config
- Attack simulation using Atomic Red Team and MITRE ATT&CK framework
- Detection engineering — custom Wazuh XML rule writing with MITRE mapping
- Incident response documentation and professional playbook writing
- Security gap analysis and remediation planning
- Alert correlation and attack chain analysis
- CIS benchmark compliance assessment
- Defense in depth validation across multiple security layers

Tools Mastered

Wazuh SIEM | Sysmon | Atomic Red Team | MITRE ATT&CK Navigator | Windows Event Logs | OpenSearch | VirtualBox | Ubuntu Server | Custom XML Detection Rules | PowerShell

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